# *PHY 317: Classical Mechanics*

*Michael G. Lerner*

*Smith College, Fall 2026*

One of the great strengths of classical mechanics is that it is the subject where physics first learned to be systematic: the same handful of ideas (Newton's laws, the conservation laws, and then the principle of least action) describe a raindrop, a satellite, a spinning top, and a chaotic pendulum. Newton's laws themselves are two courses behind you. The primary objective of this course is to take them much further than PHY 117 could – forces that depend on velocity, systems of many bodies, motion seen from accelerating and rotating frames – and then to rebuild the subject around the Lagrangian formulation, which is the language that every theory you meet after this one is written in. A significant secondary objective is to develop the fluency and the checking habits (units, limiting cases, symmetry, a numerical solution when the algebra gives out) that let you attack a mechanics problem you have never seen before with confidence. Concretely, by the end of this course you will be able to:

1. Set up and solve the equations of motion for a mechanical system in whatever coordinates the problem wants – Cartesian, polar, or your own generalized coordinates – starting from Newton's laws or from a Lagrangian.
2. Use the conservation laws (momentum, angular momentum, energy) as *tools*, and know when each one applies and why.
3. Analyze oscillations: damped, driven, resonant, coupled, and their normal modes.
4. Solve the central-force problem and read an orbit off an effective potential; work in accelerating and rotating frames.
5. Explain what chaos is and is not, and recognize it in a simulation.
6. Use Python to solve equations of motion numerically, to check analytic results, and to explore what happens when the algebra gives out.
7. Check your work the way physicists do: units, limiting cases, symmetry, and plugging the answer back in.

**Instructor:** Michael Lerner
McConnell 303
mglerner@smith.edu

**Class:** Sabin-Reed 308
Mon 1:40–2:55, Wed & Fri 1:20–2:35
First meeting Wed Sep 9
Last meeting Mon Dec 14

**Help Sessions (Office hours): TO BE SET** after the week-1 scheduling poll. I also have an open-door policy: stop in whenever my door is open. That's most of the time.

**Text:** Taylor, *Classical Mechanics* (University Science Books, 2005)

**Website:** Moodle
**Python:** `https://posit.smith.edu` – nothing to install or buy

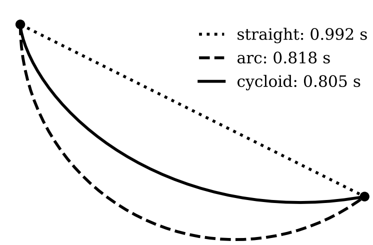


Figure 1: A bead slides without friction from the upper dot to the lower one. Which track gets it there fastest? Not the straight line. By week 8 you will be able to derive the winner (a cycloid) in about a page.

## *Prerequisites*

PHY 210 (Mathematical Methods) and PHY 215, or permission of the instructor. This course builds directly on PHY 117 and a little of PHY 118, and it uses the mathematics from PHY 210 constantly.

Just-in-time review. The course calendar on Moodle has a "Prerequisites" column that lists, for each new topic, the earlier material it leans on (by Knight chapter for intro physics and by Felder & Felder chapter for math methods). There are two good ways to use it: *just-in-time* – review those concepts before we get to the topic – or *just-after-time* – if a new idea is rough going, use the list to find and shore up the foundation right after class.[1] If you find the prerequisite material itself is the problem, come see me early; closing a gap in week 3 is easy and in week 10 it is not.

[1] This framing, and the column itself, are Will Raven's, from his Fall 2025 version of this course.

## *Course calendar*

The complete day-by-day schedule (topics, reading due, PCCIs, homework due) is posted on Moodle and kept up to date there. Read the assigned sections *before* class; class time assumes you have.

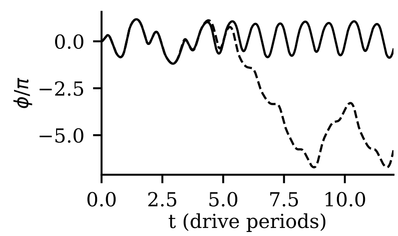


Figure 2: Two driven, damped pendulums released from angles that differ by one part in a thousand. For five drive cycles you cannot tell them apart; then one flips over the top and they never agree again. Week 14.

| Weeks | Topics | Taylor |
|---|---|---|
| 1 | Newton's laws; polar coordinates | Ch. 1 |
| 2 | Projectiles with air resistance; charged particles in magnetic fields | Ch. 2 |
| 3 | Momentum and angular momentum; rockets; center of mass | Ch. 3 |
| 4 | Energy: conservative forces, potential energy, central forces | Ch. 4 |
| 5–6 | Oscillations: damped, driven, resonance, Fourier series | Ch. 5 |
| 7–8 | Calculus of variations; Euler–Lagrange; the brachistochrone | Ch. 6 |
| 8–9 | Lagrangian mechanics: generalized coordinates, constraints, Lagrange multipliers | Ch. 7 |
| 9–10 | Two-body central-force problem; orbits | Ch. 8 |
| 11 | Non-inertial frames: tides, centrifugal and Coriolis forces | Ch. 9 |
| 12–13 | Coupled oscillators and normal modes; the double pendulum | Ch. 11 |
| 14–15 | Chaos | Ch. 12 |

**Key dates**

| | |
|---|---|
| Mon Oct 5 | Mountain Day holder |
| Mon Oct 12 | No class (recess) |
| Mon Oct 19 | Exam 1 (Ch. 2–4) |
| Fri Nov 13 | Exam 2 (Ch. 5–7) |
| Wed Nov 25 | No class |
| Fri Nov 27 | No class |
| Mon Dec 7 | Exam 3 (Ch. 8, 9, 11) |
| Wed Dec 16 | HW13 due |
| Dec 19–22 | Final exam period |

Mountain Day is announced the morning of; if it lands on a class day, that

## *How the course works*

Everything in this course is worth points, and the course total is exactly 1000 points. Within each category, each assignment is weighted equally.

| **Category** | **Number** | **Drop** | **Each** | **Total** |
|---|---|---|---|---|
| Participation & pre-class check-ins (PCCIs) | 39 | 4 | 2 | 70 |
| Weekly homework | 13 | 1 | 25 | 300 |
| Exams (in class) | 3 | 0 | 190 | 570 |
| Final exam, required part (self scheduled) | 1 | 0 | 60 | 60 |
| **Course total** | | | | **1000** |

**Grade scale** (not curved)

| | | | |
|---|---|---|---|
| A | 93+ | C+ | 77–79.9 |
| A- | 90–92.9 | C | 73–76.9 |
| B+ | 87–89.9 | C- | 70–72.9 |
| B | 83–86.9 | D+ | 67–69.9 |
| B- | 80–82.9 | D | 63–66.9 |
| | | D- | 60–62.9 |
| | | E | below 60 |

## *Pre-class check-ins (PCCIs) and attendance*

Most class days there will be a short PCCI due at the start of class, on paper: usually one Taylor problem, chosen so that working it primes you for that day's material. **You will receive full credit for a good-faith effort that includes a clear description of anything you are stuck on.** An incorrect or incomplete answer *without* such an explanation won't receive full credit.[2] A PCCI should take you less than 15 minutes; if one doesn't, please tell me. Turning in the day's PCCI is also how attendance is recorded, and your four lowest participation days are dropped, so occasional absences and off days cost you nothing.

[2] For example, if the question were to differentiate $f(g(x))$: "I don't know" earns nothing; the correct answer earns full credit; and so does "I think it's $f'(g(x))$, except I know the chain rule says to do something different because it's $f(g(x))$ and not just $x$ – I'm not clear on what, though."

This is a small class, and it runs on group work at the board. Your classmates are a resource for you and vice versa, and this in-class work is one of the main benefits of a school like Smith. If you have an accommodation, please come talk to me so that we can make the class work for both of us, but *in general*, I will expect you to miss no more than four classes, and missing more than four may adversely affect your grade.

## *Weekly homework*

Physics is learned by doing problems, and upper-level mechanics problems are long: expect a set of five or six Taylor problems each week, most of them substantial. Homework is posted on Moodle and due there **Wednesdays by 10:00 PM** as a single PDF of your written work (photos of handwritten pages are fine; combine them into one PDF). A few due dates move to sit before an exam or a break; the Moodle calendar is authoritative.[3]

[3] Most weeks one problem is **computational**, done in Python on `https://posit.smith.edu`. Include your code and its plots in the PDF. If you already know and prefer Mathematica, you may use it instead – the physics is what's graded.

You may (and should) work with others on the homework, but what you submit must represent your own understanding, written up by you. Your lowest homework score is dropped.

EVERY PROBLEM ENDS WITH AN ASSESSMENT. After the answer, write two or three lines checking it: a limiting case ($m \to 0$, $t \to \infty$, the angle going to zero), a units check, or a comparison with a result you already trust. Some problems on the sets say "include an assessment" or "this is your assessment"; the habit is expected on all of them, and it is worth points in the rubric below.

HOMEWORK IS GRADED ON CORRECTNESS, problem by problem, on a simple scale:[4]

+ 100%: complete and correct, including units and the assessment.

✓ 95%: one small problem.

− 50%: several small problems, or one big one. *Resubmitting a completely correct solution together with a few sentences of reflection raises this to 85%.*

× 25%: substantial confusion. *A correct resubmission with an in-depth reflection raises this to 75%.*

0 skipped.

A reflection says why the original attempt went wrong, what you were thinking at the time, and – most importantly – what you now understand that corrects it. Reflections and resubmissions are due within ten days of getting the homework back, on separate pages attached to the next set. A reflection without real engagement earns less.

[4] The rubric and the reflection option are Will Raven's, kept because they work: the students who wrote reflections were the ones whose exam scores moved.

## *Exams*

Three exams, in class, taking the full period, on the dates in the margin above. Each exam has **three problems, one per chapter covered**; a student with command of the material should finish each in about 25 minutes. Exams are closed-book; you may bring **one 8.5×11 sheet of handwritten notes, both sides**.[5] Exam problems are closely aligned with the in-class problems and the homework, and each exam covers only the chapters listed – never the week the exam is given.

[5] You may use a basic calculator; no phones, laptops, or other internet-connected devices. Students with documented accommodations will, of course, have those accommodations met.

## *Final exam, with redemption problems*

The final, during the December 19–22 self-scheduled exam period, has two parts:

1. **Required** (60 points): one problem on Chapter 12 (chaos), the only chapter not covered by an in-class exam.
2. **Optional grade-replacement problems**: nine more problems, one for each chapter that appeared on the three exams (Chapters 2–9 and 11). Attempt any of them. For each one you solve, the new score *replaces* your score on the corresponding exam problem if and only if it is higher. There is no penalty for an attempt; this can only improve your grade.

Do the required problem first; then do as many redemption problems as you want and have time for. I keep per-problem exam scores all semester precisely so this works.

### *Partial credit, and how to get it*

Several of the problems in this class are quite challenging. For the harder problems, my goal is to have you make the strongest possible effort toward **understanding** the solution. Thus, if you cannot fully solve a problem, say whatever you can about the way in which a solution would proceed from where you stop; say whatever you can about the qualitative behavior of a solution; say whatever you can about the physical meaning of the solution. And always, always: **check your units**, **check the limiting cases**, and **plug your solution back into the equation of motion**.[6]

[6] You are never done solving an equation of motion until you have verified that your solution works.

### *Late policy*

It is extremely hard to catch up on late work (usually things are late because you are busy, and people don't tend to get less busy!). That is why every category has built-in drops: four participation days, one full homework set. My strong suggestion, when life happens, is to use the drops and prioritize getting your schedule back under control.

Beyond the drops: all students start the term with **3 free late passes**. You can spend a late pass to extend a homework set to the following class period with no penalty. Late passes are not automatic – email me before the deadline to say you're using one, and once declared it is spent whether or not you turn the work in by the extended date.[7]

[7] Late passes cannot be applied to PCCIs (their whole value is priming you for that day's class – the drops cover them), to exams, or to the final. After your free passes are spent, further extensions cost 10%, then 20%, and so on, of that assignment's points.

## *Python (and Mathematica)*

Computation is part of doing mechanics: the equations of motion for quadratic drag, the driven pendulum, and the double pendulum have no closed-form solutions, and numerical solutions are how physicists actually see those systems. We will use Python (NumPy, SciPy, Matplotlib) in Jupyter notebooks on `https://posit.smith.edu`, the same setup as PHY 210. I'll post starter notebooks for the computational homework problems and for the in-class simulations. Two expectations, borrowed from Will Raven's Mathematica policy for this course:

1. **Simplify first.** Anything that can be done with physics – an integral that vanishes by symmetry, a limit you can read off – gets done on paper first. The computer confirms; it does not replace your understanding.
2. **Explore and assess.** Use the computer to push past the algebra: vary a parameter, plot the thing, check a limiting case numerically. Say in your write-up what you predicted *before* running it and what you concluded *after*.

Previous offerings used Mathematica, and Smith has a license; if you already know it and prefer it, you may use it for any computational problem.

## *A short statement on inclusion and equity*

I can hardly overstate how much inclusion, equity and diversity considerations have shaped my entire thinking around teaching and practicing physics. It affects everything from the way that physics is framed historically to the way that it is typically taught in your classrooms. My main goal is to create an environment that tells all of us that we can do physics; that invites people into physics who have never been invited, or who have been explicitly excluded.

Physics cannot be divorced from the world in which we study it. I expect that we will recognize that in this class, and I expect that we will create an inclusive and welcoming environment. More specifically, I expect that we will strive *not* to create an environment that excludes people. This is hard work. We will get things wrong. You will get things wrong. *I* will get things wrong. Especially when I get things wrong, I encourage you to tell me, using any of the channels under *Sensitive issues* below.

## *Use of artificial intelligence*

Before we get to a policy, two things:

- AI models make subtle errors, hallucinate, and hand you technically-correct solutions that teach you nothing; your job here is to verify, question, and ultimately understand the physics.

- Your PCCIs are graded on the basis of "good-faith effort", *not* correctness – the only thing I'm grading is a record of *your* thinking, explicitly including credit for starting and then saying where you got stuck. Having AI polish that doesn't help your grade at all, and cheats you of learning. And the homework reflections only work if they are honestly yours.

So, the policy: AI is a powerful tool, and it offers real potential to enhance your learning in this course. Used improperly, it can genuinely hinder your development as a physicist. You are welcome to use AI in this class, provided you always adhere to the Academic Integrity Statement: **using AI to solve homework, PCCI, or exam problems is a violation of the Academic Integrity policy and will be treated as such**, and any AI use in work you submit must be cited, like any other source.[8]

We're happy to use AI where it actually helps in this class, but we have to be careful not to use it before then. So, feel free to use AI to dig further in, to get alternate explanations, to check rote algebra after you've done it, to debug code by having it explain its diagnostic process rather than hand you the fix, to explore new topics, etc. But keep it in check to make sure you're in charge of your own learning. Early in the semester, we will spend part of a class deciding *together* how AI should and shouldn't fit into homework in this course – computational work especially – and we'll post what we decide on Moodle next to this policy.

[8] Adapted, with thanks, from Will Raven, whose "AI prompts for students" handout (on Moodle) is a good guide to using it *well*: asking it to quiz you, to critique a reflection draft, or to pose "what if" variations of a problem you have already solved.

## *Policies and logistics*

Communication. My preferred mode of communication is email. I will check my email multiple times per day during the work week. If you have sent me an email and have not heard back from me within 24 hours (excluding weekends), you should email me again or drop by my office to ask your question in person. I cannot guarantee that I will regularly check my email on evenings or on weekends.[9]

[9] I expect that you will check your Smith email at least once per day: 24 hours after I email the class, I expect everyone has read it. Emailed assignments, due dates, and policy changes carry the same weight as this syllabus, and the same goes for Moodle.

Workload. This course should take about 12 hours of your time per week: about 4 in class, and about 8 outside it on reading, PCCIs, homework problems, and review. **If you find yourself spending significantly more (or less) than this, please let me know ASAP! That's an instructor-error that I can easily fix.**

Sensitive issues. What if you have something you want to talk about, but don't feel comfortable doing so in a public (or non-anonymous) space? This often comes up with issues of racism, sexism, and other kinds of discrimination. It can come up with issues that relate to my behavior, the behavior of your classmates, or the world in general. Please feel *encouraged* to use either the anonymous feedback form linked on Moodle, or to leave a written note under my door.[10] Of course, if you feel comfortable and safe doing so, you are also encouraged to talk publicly or non-anonymously about these things! If you are not comfortable with any of these options, the Resources section below lists offices that can help, including confidential ones.

[10] If you want to write a note, you may want to type it up and print it out so that your handwriting is further anonymized.

RECORDING. My plan is to record from my iPad to both PDF and videos, posted to Moodle. We'll see how technology works.[11] Sometimes the mic picks up conversations in the room. If that's a problem, I can look into muting it. Let me know.

[11] I do this because several students have used it as a post-class resource in the past. Sometimes a derivation is worth reviewing, sometimes you were having a spacey day, etc. But it's worth knowing that student experience shows that it's a very poor substitute for attendance, and just watching the recordings won't count towards your 70 attendance/participation points.

ACADEMIC INTEGRITY. Real learning happens when students conduct themselves with academic integrity, and our goal here is to provide you with the opportunity for (and expectation of) that learning. Read the Academic Integrity Statement. That statement is a gift to the community, giving you all sorts of privileges (self-scheduled finals, the expectation that I treat you like academic adults, etc.). And it comes with responsibilities.[12]

[12] In this class specifically: you are encouraged to work together on the homework and to study together for exams, but PCCIs should be attempted individually first, everything you turn in must be your own work, and exams and the final must be done individually.

ACADEMIC ACCOMMODATIONS. If you need classroom accommodations, please contact the Accessibility Resource Center and file things through the Accommodate system. I promise to be flexible in meeting any documented need for accommodations in a way that works well for you. If you would like some accommodation but do not have an ARC reference, let me know and we can discuss your concerns: regardless, I want to provide you the support you need to succeed in this course, though I will do so in a way that does not diminish the accommodations of others. Remember to contact me well in advance of when you need the accommodations in place, so that we can prepare.

COURSE FEEDBACK. There will be an anonymous mid-semester feedback form on Moodle around fall break, and during the last week of classes approximately 15 minutes at the beginning of one class will be set aside for you to complete the official course feedback questionnaire. Please don't wait until then if something isn't working.

## *Recommended books, and other resources*

BOOKS. Taylor is unusually readable; most weeks the reading is the best preparation you can do. When you want a second voice: Morin, *Introduction to Classical Mechanics*, has the best problem collection in the subject (with solutions) and a more demanding style; Kleppner & Kolenkow, *An Introduction to Mechanics*, is the classic at a slightly lower level and is excellent on rotating frames and angular momentum; Marion & Thornton, *Classical Dynamics of Particles and Systems*, is the traditional alternative to Taylor and covers the same ground in a different order. *The Feynman Lectures on Physics, Volume I* (free at feynmanlectures.caltech.edu) is worth reading on the principle of least action (Ch. 19 of Volume II, actually) once we reach Lagrangians. Later in your career: Goldstein, *Classical Mechanics*, the graduate standard.

PEOPLE AND PLACES. There are many resources at Smith to help you succeed in this course and in your college career:

- Your instructor! I am here to help you. That's my job, so don't think it's bothering me to have you reach out with questions or concerns.
- **[TODO: course tutor / physics help-room hours, once scheduled.]**
- The Spinelli Center – open hours, one-on-one appointments, and math review workshops.
- Writing help, learning-specialist services, and workshops on time management and managing stress, all from the Jacobson Center.
- Crisis resources, counseling, and wellness services at the Schacht Center.
- Resources for gender identity and expression, and the Office for Equity and Inclusion in general.
- Where to report sexual misconduct and other discrimination. Please note that your instructors are Responsible Reporters.